

LegalEase

Phase 5: Project Development

AI-Powered Legal Document Drafting and Export Platform

Project Phase Document

This PDF consolidates the prepared project materials for this phase.

README

Phase 5 — Project Development

5.1 Development Sequence

Step 1 — Create environment

```
python -m venv .venv
```

Windows:

```
.venv\Scripts\activate
```

Install dependencies:

```
pip install -r requirements.txt
```

Step 2 — Configure environment

Create .env:

```
GEMINI_API_KEY=your_api_key_here  
BACKEND_URL=http://127.0.0.1:8000
```

Never commit the real key.

Step 3 — Start FastAPI

```
uvicorn backend.main:app --reload
```

API:

```
http://127.0.0.1:8000
```

Step 4 — Start Streamlit

Open another terminal:

```
streamlit run frontend/app.py
```

Step 5 — Test health

Open:

```
http://127.0.0.1:8000/health
```

Expected:

```
{"status": "ok"}
```

5.2 Development Rules

- Keep API schemas in backend/schemas.py.
- Keep routes thin.
- Keep Gemini-specific code in backend/ai_core/.
- Keep document formatting in exporter modules.
- Do not place API keys in source code.
- Validate all external input.
- Log useful errors without exposing secrets.

5.3 Suggested Module Responsibilities

```
backend/  
■■■ main.py # FastAPI app
```

```
■■■ routes.py          # API endpoints
■■■ schemas.py         # Request/response validation
■■■ ai_core/
    ■■■ gemini_generator.py

frontend/
■■■ app.py             # Streamlit UI
■■■ exporters.py       # TXT/DOCX/PDF generation
```

02 API Implementation

API Implementation Specification

Health Endpoint

```
@app.get("/health")
def health():
    return {"status": "ok"}
```

Generation Endpoint

```
@app.post("/generate")
def generate(request: DocumentRequest):
    document = generate_document(request)
    return {
        "success": True,
        "document": document
    }
```

Validation Model

```
class DocumentRequest(BaseModel):
    document_type: str
    party_a: str
    party_b: str
    effective_date: date
    terms: str
    additional_instructions: str = ""
```

Error Response

Use a consistent structure:

```
{
    "success": false,
    "error": "Readable error message"
}
```

Do not expose API keys, stack traces, or internal secrets to users.

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